

MEDICARE HOSPITAL VALUE ANALYSIS

What makes a high-value hospital?

CMS star ratings don't tell the whole story.

SQL, Python, Tableau, Databricks · 7 public CMS datasets · ~5,000 U.S. hospitals

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github.com/KabiraL/Hospital_Assessment_Databricks



Which hospitals deliver higher risk-adjusted outcomes at lower risk-adjusted cost? What sets them apart?

You don't get what you pay for.

Outcomes and cost are only weakly linked, so the best-value hospitals (strong outcomes at low cost) are efficient **exceptions** and not the rule.

Stars miss hidden value.

CMS star ratings ignore cost, so **2 in 5** high-value hospitals sit at just 1–3 stars.

You don't get what you pay for.

Higher cost buys only **marginally** better outcomes.

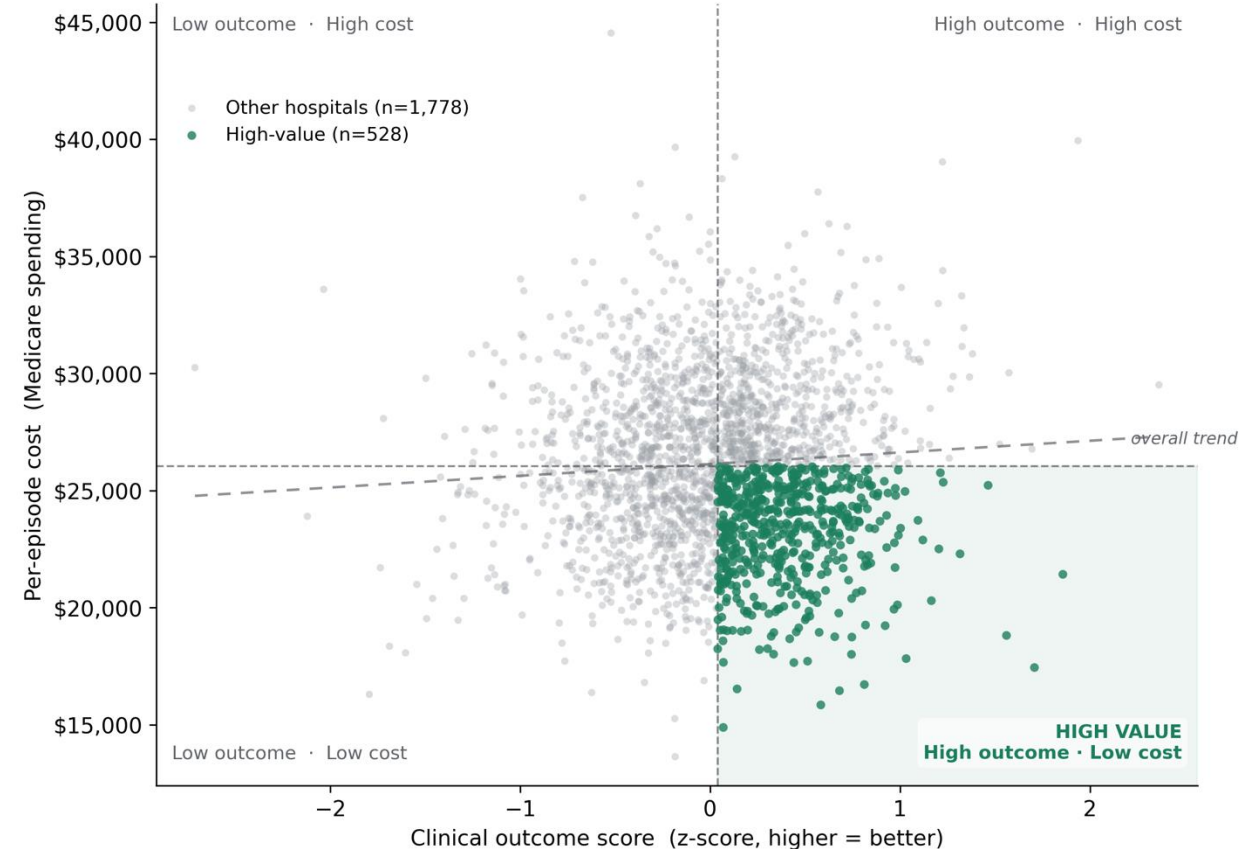
Because the two are only weakly linked, the genuine high-value hospitals (strong outcomes at lower cost) are off-trend exceptions, clustered in the lower-right.

528

high-value hospitals with above-median outcomes at below-median cost.

U.S. hospitals by clinical outcome and cost

The high-value group (high outcomes at low cost) is a small, off-trend slice.



Each dot is a hospital, plotted by clinical risk-adjusted outcome and cost.

Analysis set: 2,306 hospitals that have a CMS star rating, clinical outcomes data, and Medicare cost data (from an original set of ~ 5,000 hospitals).

Outcomes: risk-adjusted mortality, safety, and readmission.

Cost: risk-adjusted Medicare spending per episode (MSPB).

Three findings.

1

Cheap signals predict cost, but not outcomes.

Operations, patient experience, size, ownership, and location track moderately with cost but barely explain outcomes. Happier patients don't necessarily have better clinical outcomes.

2

Stars see quality, not value.

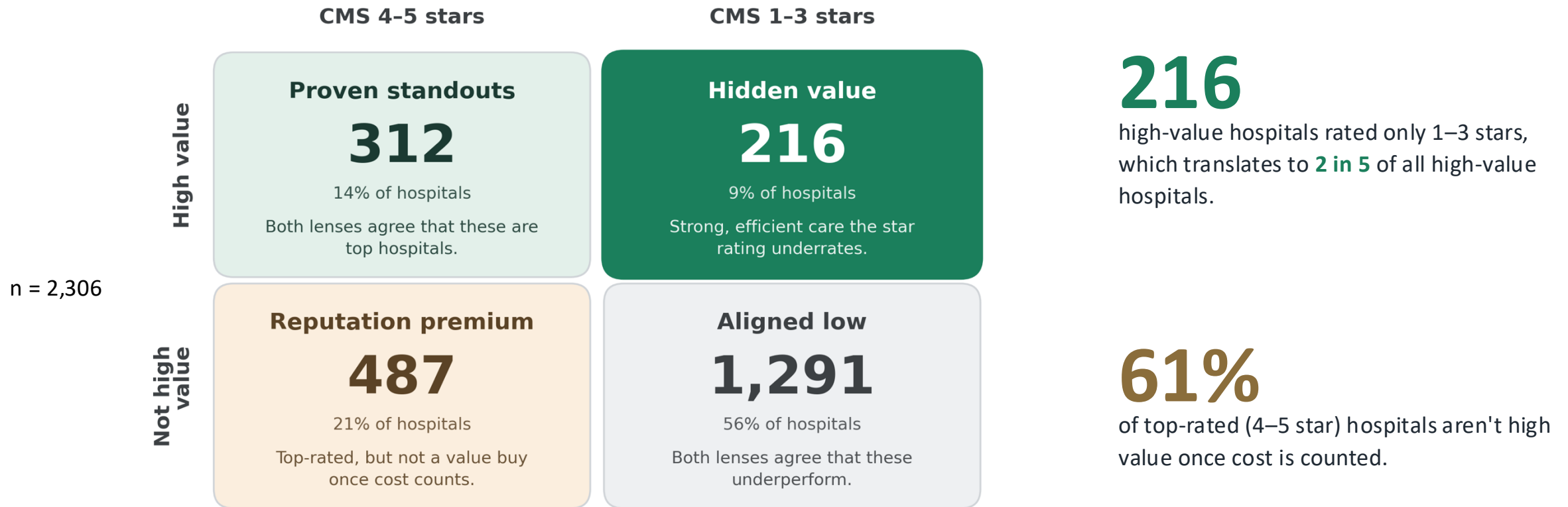
My outcome score is built from the same clinical measures the star rating relies on so stars and value largely agree on quality. The differences are almost entirely about cost, which is the single dimension from my analysis that stars structurally can't see.

3

The best hospitals fight the grain.

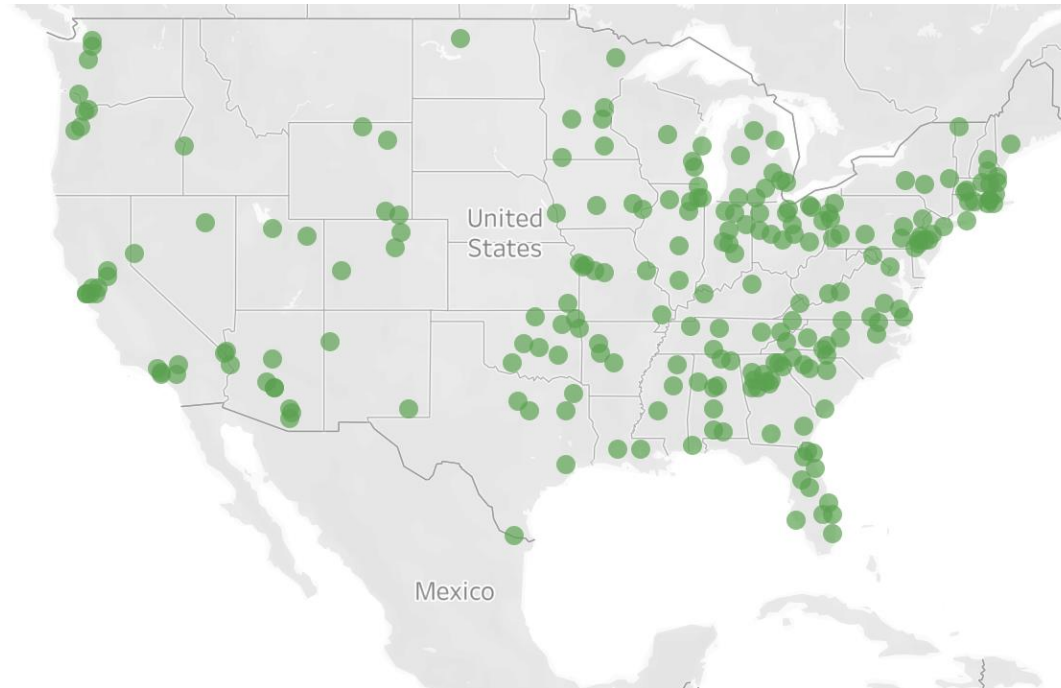
Outcomes and cost are only mildly linked, so high-outcome / low-cost hospitals are genuine **efficiency outliers**. Coarse proxies cannot reliably find them.

Stars can't see **cost**, so they miss some of the **value**.



Stars are a fair **quality** screen, but not a **value** screen. Most 1-3 star hospitals (56%) still aren't high value. The hidden value is the **efficient minority** you surface only by considering **cost**.

Where the high-value hidden hospitals are.



Value tilts modestly by region and ownership

New England and non-profit systems lean favorable; some Southern divisions and government-run hospitals lean the other way. But variation within every group is large. Region and ownership are more cheap proxies. Like size and reviews, they nudge the odds, but they cannot reliably find value. The map shows where to look for underrated value, but it cannot give a **guarantee**.

Value must be **measured**. Cost can be **screened**. Build the screen but measure the rest.

For payers, health systems, and care-coordination teams managing total cost of care, true high-value providers can't be inferred from price, reviews, or reputation. None of these reliably mirrors clinical quality. But the cost half can be screened quickly and reliably, and the hidden-value hospitals are exactly the **efficient** providers a star-only screen would miss.

Measure outcomes directly.

Don't let low cost, strong reviews, reputation, or region stand in as a proxy for clinical quality. Outcomes must be measured and cannot be inferred.

Layer cost as the efficiency screen.

Risk-adjusted Medicare per-episode spending is cheap to pull, and it's the dimension star ratings omit. Putting it next to an outcomes view is what surfaces value.

Look past the stars.

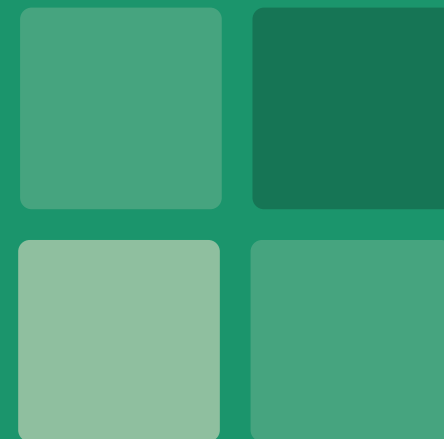
A star-only screen overlooks 216 efficient providers. This means **2 in 5** high-value hospitals are missed. The value flag catches them.

A reproducible process: Use 7 public CMS datasets → Assign a value flag per hospital → Refresh on CMS's annual cadence → Use for network tiering, referrals, and provider scorecards.

Cautions: Cost reflects Medicare spend only; the process flags candidates and not guarantees; region, ownership and other measures provide context and not filters.

TAKEAWAY

**Higher spending doesn't reliably buy better outcomes.
And a star-only screen overlooks 2 in 5 of the most
efficient hospitals.**



The fix is buildable: measure outcomes, screen cost, refresh on the annual CMS cadence.

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